Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
BCA | Sem-3 | Summer-2023

Course Code	:	2104CS301	**Date**	:	24-04-2023
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define a computer network with its characteristics.	**4**
	(B)	List the applications of Computer networks.	**3**
		OR	
		List the advantages of Computer Network.	
	(C)	Define Network Topology with an example.	**7**
		OR	
		Define Circuit switching & Packet switching with Pros and Cons.	
Q.2	**(A)**	Describe Data-Link Layer.	**4**
	(B)	List the layers in OSI Reference Model.	**3**
		OR	
		Draw the OSI Reference Model.	
	(C)	Explain TCP/IP Model with diagram.	**7**
		OR	
		Explain functions of Transport Layer.	
Q.3	**(A)**	Describe twisted pair cables with their types.	**4**
	(B)	Define Transmission Media. Classify Transmission Media.	**3**
		OR	
		Explain Infrared with an example.	
	(C)	Explain Bluetooth Architecture with a diagram.	**7**
		OR	
		Classify microwave communication with their types.	

Q.4	**(A)**	Compare hub and switch.	**4**
	(B)	Define networking device. List the devices used in networking.	**3**
		OR	
		Write a short note on Repeater.	
	(C)	Carry out functionality of router in networking.	**7**
		OR	
		Carry out functionality of switch in networking.	
Q.5	**(A)**	List IP address Classes and range of each class.	**4**
	(B)	Write a short note on DNS.	**3**
		OR	
		Define Web and HTTP.	
	(C)	Given IP address 192.168.1.10/27. Find below details: a) Class b) Subnet mask c) What are the total subnets? d) What are the total hosts?	**7**
		OR	
		Given IP address 172.16.0.0/25. Find below details: a) Class b) Subnet mask c) What are the total subnets? d) Give the first host ID and the last host ID of each subnet.	

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